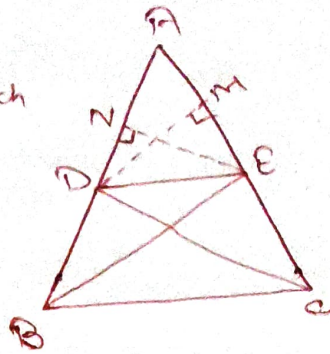


# Thales Theorem {BPT}



Use this method  
for 5 marks  
only

Given:- A  $\triangle ABC$  in which  
 $DE \parallel BC$



To prove:-  $\frac{AD}{BD} = \frac{AE}{EC}$

Proof:-

Construction:-

- Draw a line from D on AE
- Draw a line from E on AD
- Join BE & DC

In  $\triangle ADE$  as Base AE

$$\text{Area of triangle} = \frac{1}{2} \times AE \times DM \quad \text{--- (i)}$$

In  $\triangle ADE$  as Base AD

$$\text{Area of triangle} = \frac{1}{2} \times AD \times EN \quad \text{--- (ii)}$$

In  $\triangle DBE$

$$\text{Area of triangle} = \frac{1}{2} \times DB \times EN \quad \text{--- (iii)}$$

In  $\triangle DEC$

$$\text{Area of triangle} = \frac{1}{2} \times EC \times DM \quad \text{--- (iv)}$$

Ratio of eq<sup>n</sup> (ii) & (iii) <sup>will</sup> is always equal to (i) & (iv)

$$\frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN}$$

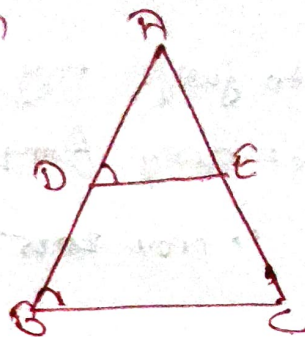
$$\frac{AE}{EC} = \frac{AD}{BD}$$

Hence Proved

Agushy  
06/01/20

Given:  $\triangle ABC$  is a triangle in which  $DE \parallel BC$

To Prove:-  $\frac{AD}{DB} = \frac{AE}{EC}$



Use this method for 3 marks

Proof:-

In  $\triangle ADE$  &  $\triangle ABC$

$\angle ADE = \angle ABC$  {Corresponding angles}

$\angle DAE = \angle BAC$  {Common Angle}

$\triangle ADE \sim \triangle ABC$  {A.A Similarity}

$$\frac{AD}{AB} = \frac{DE}{BC} = \frac{AE}{AC}$$

$$\frac{AD}{AB} = \frac{AE}{AC}$$

$$\frac{AB}{AD} = \frac{AC}{AE} \quad \text{{Reciprocal}}$$

$$\frac{AB}{AD} - 1 = \frac{AC}{AE} - 1$$

$$\frac{AB - AD}{AD} = \frac{AC - AE}{AE}$$

$$\frac{BD}{AD} = \frac{EC}{AE}$$

Again reciprocal

$$\frac{AD}{BD} = \frac{AE}{EC}$$

Proved

~~08/01/25~~